Statistical Methods for Biomedical Data Science

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Course Content

1. Data types, summary statistics, and graphical presentation of data

2. Introduction to Probability and Statistical Theory

Data Types

Data types refer to the kinds of values a variable can take. They are broadly classified as quantitative or qualitative.

Qualitative

Qualitative data, also known as categorical data, are data that can be placed into distinct categories, according to some characteristic or attribute. E.g. lables or names

Quantitative

Quantitative data are numerical and can be ordered or ranked. E.g Number of children, height, weight.

Data Types Cont'd

Quantitative data can be further classified into two groups: discrete and continuous.

- **Discrete variables** assumes values that can be counted.
- **Continuous variables** can assume an infinite number of values between any two specific values.

In addition to being classified as qualitative or quantitative, variables can be classified by how they are categorized, counted, or measured. This type of classification uses **measurement scales** and four common types of scales are used: nominal, ordinal, interval, and ratio.

Data Types Cont'd

Nominal

This level of measurement classifies data into nonoverlapping categories in which no order or ranking can be imposed. e.g. political party (NPP, NDC, CPP) and religion (Christianity, Islam, Judaism)

Ordinal

The ordinal level of measurement classifies data into categories that can be ranked; however, precise differences between the ranks do not exist.

Interval

The interval level of measurement ranks data, and precise difference between units of measure do exist. However, there is no meaningful zero. IQ is an example of such variable. Temperature is another example

Data Types Cont'd

Ratio

This level of measurement possesses all the characteristics of the interval measurement, and there exists a true zero. Examples of ratio scales are those used to measure height, weight, area, and number of phone calls received.

Summary Statistics

Summary statistics provide a quick overview of data by describing key features like central tendency (mean, median), variability (range, standard deviation), and the shape of the distribution. They help simplify and interpret large datasets.

Measures of Central Tendency

These statistics describe the center or typical value of a dataset. They help identify where most values in a dataset tend to cluster: mean, median, and mode.

- **Mean**: The mean is the sum of the values, divided by the total number of values. The symbol $\overline{x}$ represents the sample mean.

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

Summary Statistics Cont'd

- **Median:** The middle value when the data is ordered.
- **Mode:** The most frequently occurring value.

Measures of Dispersion

These describe the spread of variability in a dataset.

- **Range**: The range is the difference between the highest and lowest value.
- **Variance**: The population variance is the average of the squares of the distance each value is from the mean. It is computed as

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2$$

- **Standard Deviation**: The standard deviation is the square root of the variance.

Summary Statistics Cont'd

Measures of Position

The measures of position are used to locate the relative position of a data value in a data set.

- **Percentiles**: They divide the data set into 100 equal groups.
- **Quartiles**: They divide the distribution into four groups seperated by Q_1, Q_2, Q_3

Measures of Shape

These describe the symmetry and peakedness of a distribution

- **Skewness**: Indicates whether or not the distribution is symmetric
- **Kurtosis**: Shows or reflects the heaviness of the tails

Graphical Presentation of Data

Graphical presentation uses visual tools to explore and communicate data patterns, trends, and distributions clearly and effectively. They include Histogram, Stem and leaf plot, pie chart etc.

Histogram

The histogram is a graph that displays the data by using contiguous vertical bars (unless the frequency of a class is 0) of various heights to represent the frequencies of the classes.

Pie Chart

A circular graph divided into slices to illustrate proportions or percentages of a whole.

Graphical representations of the histogram and pie chart are shown below:

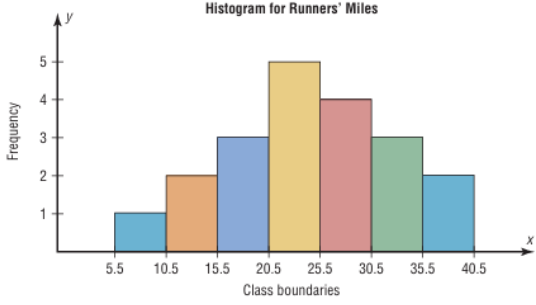


Figure: Histogram

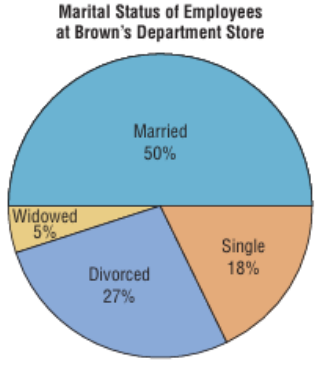


Figure: Pie Chart

Introduction to Probability

Probability

Probability as a general concept can be defined as the chance of an event occurring. Many people are familiar with probability from observing or playing games of chance, such as card games, slot machines, or lotteries. In addition to being used in games of chance, probability theory is used in the fieldsofinsurance,investments,and weather fore casting and in various other areas.

Key Terms

- **Experiment:** A process with uncertain outcome
- **Outcome:** A result of a single trial of a probability experiment.
- **Sample Space:** A sample space is the set of all possible outcomes of a probability experiment.
- **Event:** A subset of the sample space

Basic rules of probability

There are four basic probability rules. These rules are helpful in solving probability problems, in understanding the nature of probability, and in deciding if your answers to the problems are correct.

- **Rule 1:** The probability of any event, E, ranges from 0 to 1. i.e $0 \leq P(E) \leq 1$
- **Rule 2:** If an event cannot occur, it's probability is 0.
- **Rule 3:** If an event E is certain, then the probability of E is 1.
- **Rule 4:** The sum of the probabilities of all the outcomes in the sample space is 1

Additional Rules

Mutually exclusive events: Two events A and B are mutually exclusive if $P(A \cap B) = 0$

Complementary events: If the probability of an event or the probability of its complement is known, then the other can be found by subtracting the probability from 1. $P(E) = 1 - P(\overline{E})$

Types of Probability

Classical Probability

Classical probability uses sample spaces to determine the numerical probability that an event will happen. Classical probability assumes that all outcomes in the sample space are equally likely to occur.

Under classical probability, the probability of any event, E, is given by

$$P(E) = \frac{n(E)}{n(S)}$$

Empirical Probability

Unlike classical probability which assumes that outcomes are equally likely, empirical probability relies on actual experience to determine the likelihood of outcomes.

Given a frequency distribution, the probability of an event being in a given class is

$$P(E) = \frac{\text{frequency for the class}}{\text{total frequencies in the distribution}} = \frac{f}{n}$$

This probability is called empirical probability and is based on observation.

Subjective Probability

This type of probability uses a probability value based on an educated guess or estimate, employing opinions and inexact information. For example, a sportswriter may say that there is a 30% probability that Ghana will win the world cup next year.

The Multiplication Rules

The **multiplication rules** can be used to find the probability of two or more events that occur in sequence.

Note: Two events A and B are **independent events** if event A occurring does not affect the probability of event B occurring.

Rule 1:

When two events are independent, the probability of both occuring is

$$P(A \text{ and } B) = P(A) \cdot P(B)$$

Note: When the outcome of one event changes the probability of the occurrence of another, the events are said to be **dependent events**

Rule 2: When two events are dependent, the probability of both occurring is

$$P(A \text{ and } B) = P(A) \cdot P(B|A)$$

Example

A bag contains 5 red marbles and 3 blue marbles.
(a) If one marble is drawn at random, with replacement, and then a second marble is drawn, what is the probability that both marbles are red?
(b) If one marble is drawn at random, without replacement, and then a second marble is drawn, what is the probability that both marbles are red?

Solution

Let R be the event of selecting a red marble and B, a blue marble.

$P(R) = \frac{5}{8}$ and $P(B) = \frac{3}{8}$

(a) $P(\text{selecting two red balls}) = \frac{5}{8} \cdot \frac{5}{8} = \frac{25}{64}$

(b) $P(\text{both marbles being red}) = \frac{5}{8} \cdot \frac{4}{7} = \frac{5}{14}$

Conditional Probability

The **conditional probability** of an event B in relationship to an event A is the probability that event B occurs after event A has already occurred. The formula is

$$P(B|A) = \frac{P(A \text{ and } B)}{P(A)}$$

Example

A box contains black chips and white chips. A person selects two chips without replacement. The probability of selecting a black chip and a white chip is $\frac{15}{56}$, and the probability of selecting a black chip on the first draw is $\frac{3}{8}$. Given that the first chip selected was black, find the probability of selecting a white chip on the second draw.

Solution

Let B be the event of selecting a black chip and W be the event of selecting a white chip.
$P(B \cap W) = \frac{15}{56}$, $P(B) = \frac{3}{8}$
Then,

$$P(W|B) = \frac{P(B \text{ and } W)}{P(B)} = \frac{\frac{15}{56}}{\frac{3}{8}} = \frac{5}{7}$$

Trial Questions

1. A fair six-sided die is rolled twice. What is the probability of getting a 4 on the first roll AND a 5 on the second roll? **Ans: 1/36**

2. A box contains 10 light bulbs, of which 3 are defective. Two light bulbs are selected at random without replacement. What is the probability that the first bulb selected is defective AND the second bulb selected is not defective? **Ans: 7/30**

3. In a certain city, 60% of the population owns a car, and 25% of the population owns a motorcycle. 15% of the population owns both a car and a motorcycle. If a person is chosen at random and is found to own a car, what is the probability that they also own a motorcycle? **Ans: 0.25**

The Fundamental Counting Principle

If there are 'm' ways to do one task and 'n' ways to do a second task, then there are m x n ways to perform both tasks in sequence. This principle can be extended to more than two tasks. In a sequence of n events in which the first one has k_1 possibilities and the second event has k_2 and the third has k_3, and so forth, the total number of possibilities of the sequence will be

$$k_1 \cdot k_2 \cdot k_3 \cdots k_n$$

Example: A man has 3 shirts, 4 trousers, and 2 shoes. How many different outfits can the man make?

Solution

3 shirts x **4** trousers x **2** shoes = **24** different outfits

Permutation

A permutation is an arrangement of objects in a specific order. The arrangement of n objects in a specific order using r objects at a time is written as ${}^nP_r$ and the formula is

$$ {}^nP_r = \frac{n!}{(n-r)!} $$

Example 1: Ten people enter a contest. In how many ways can 1st, 2nd, and 3rd place be determined?

Solution

n = 10 r = 3

No. of ways = $\frac{10!}{(10-3)!} = 720$

Example 2 (Permutation with repeats): In how many ways can the letters in the word MISSISSIPPI be arranged?

Solution

Notice there are 11 letters total but the S and I each occur four times and the P occurs twice. Hence,

No. of arrangement $= \frac{11!}{4!4!2!} = 34650$

Example 3: In how many different ways can 5 people arrange themselves around a circular table?

Solution

(5-1)! $= 24$

Combination

Combinations are used when the order or arrangement is not important. The number of combinations of r objects selected from n objects is denoted by ${}^nC_r$ or $\binom{n}{r}$. The formula is

$$ {}^nC_r = \frac{n!}{r!(n-r)!} $$

Example 1: From a group of 5 friends, you must select 3 friends to join you on a camping trip. In how many different ways can you select three friends?

Solution

No. of ways $= {}^5C_3 = \frac{5!}{3!(5-3)!} = 10$

Example 2: From a group of 5 women and 7 men, how many different committees consisting of 2 women and 3 men can be formed?

Solution

Hint: Find the number of combinations for each group separately and then multiply them together.

Group 1 x Group 2

$$= \binom{5}{2} \cdot \binom{7}{3}$$

$$= \frac{5!}{2!(5-2)!} \cdot \frac{7!}{3!(7-3)!}$$

$$= 10 \cdot 35$$

$$= 350$$

Thank You